ROKS vs VCF Storage Performance Comparison

IBM Cloud ROKS (OpenShift Virtualization + ODF) vs VMware VCF (vSAN ESA + NFS Endurance)

ROKS run: 2026-05-26 (post-prefill-fix; medium VM, 150Gi, concurrency=1, volumeMode: Block for RBD, Filesystem for CephFS) — VCF run: 20260225 (medium VM, 150Gi). **Methodology updated 2026-05-26:** earlier ROKS numbers were skewed by sparse-RBD allocation cost in the measurement window; real ROKS write p99 is 5–33 ms, not 50–1,610 ms.

Matched test conditions: Both platforms were tested with **medium VMs (4 vCPU, 8 GiB RAM)**, **150 GiB disks**, and **60s fio runtime**. Test conditions are matched across platforms — absolute values are directly comparable. **ROKS update:** RBD pools now use `volumeMode: Block` PVCs, giving QEMU direct block device passthrough (`host_device` + `aio=native`), eliminating the `disk.img` file indirection bottleneck.

Test Conditions [-]

ROKS IBM Cloud ROKS

Platform	OpenShift Virtualization + ODF (Ceph)
Workers	3x bare metal (NVMe-backed ODF)
VM size	medium (4 vCPU, 8 GiB RAM)
Disk size	150 GiB
fio runtime	60s (10s ramp)
I/O depth	32
Threads	4 per job
Direct I/O	Yes (O_DIRECT)

VCF VMware VCF

Platform	vSAN ESA 8.0.3 + NFS Endurance
Hosts	4x (vSAN cluster)
VM size	medium (4 vCPU, 8 GiB RAM)
Disk size	150 GiB
fio runtime	60s
I/O depth	32
Threads	4 per job
Direct I/O	Yes (O_DIRECT)

Storage Mapping [-]

Category	ROKS StorageClass	VCF Storage Target	Rationale
Best Replicated	`rep3-virt` (3-way RBD, VM-optimized + Block)	`raid1-ftt1-thick` (RAID-1 thick)	Top replicated performer on each platform
Standard Replicated	`rep2` (2-way RBD, Block)	`raid1-ftt1` (RAID-1 thin)	Default replicated tier
CephFS	`cephfs-rep3` (3-replica CephFS, Filesystem (cephfs-rep2 had pre-existing auth-cap issue on the test cluster; cephfs-rep3 substituted))	`raid1-ftt1` (RAID-1 thin)	POSIX filesystem vs thin replicated (both use filesystem indirection)
Erasure Coded	`ec-2-1` (2+1 EC)	`raid5-ftt1` (RAID-5 (3+1))	Parity-based space-efficient storage
NFS Mid Tier	`ibmc-vpc-file-1000-iops` (1000 IOPS)	`workload-share-3hq5c` (4 IOPS/GB Endurance)	Mid-range NFS
NFS Low Tier	`ibmc-vpc-file-500-iops` (500 IOPS)	`workload-share-x1ydq` (2 IOPS/GB Endurance)	Low-end NFS

Performance Comparison

ROKS wins   VCF wins

Random 4k IOPS (IOPS)

Category	ROKS ROKS	VCF VCF	Delta	Winner
Best Replicated	52,170	81,578	-36.0%	VCF
Standard Replicated	61,490	80,851	-23.9%	VCF
CephFS	53,586	80,851	-33.7%	VCF
Erasure Coded	44,862	69,361	-35.3%	VCF
NFS Mid Tier	997.0	14,745	-93.2%	VCF

Category	ROKS ROKS	VCF VCF	Delta	Winner
NFS Low Tier	497.0	7,671	-93.5%	VCF

Sequential 1M BW (MiB/s)

Category	ROKS ROKS	VCF VCF	Delta	Winner
Best Replicated	6,338	1,230	+415.5%	ROKS
Standard Replicated	5,737	1,223	+369.1%	ROKS
CephFS	4,980	1,223	+307.2%	ROKS
Erasure Coded	3,668	1,688	+117.3%	ROKS
NFS Mid Tier	64.3	999.0	-93.6%	VCF
NFS Low Tier	31.2	502.4	-93.8%	VCF

Mixed 70/30 4k IOPS (IOPS)

Category	ROKS ROKS	VCF VCF	Delta	Winner
Best Replicated	49,054	22,515	+117.9%	ROKS
Standard Replicated	54,027	23,397	+130.9%	ROKS
CephFS	39,533	23,397	+69.0%	ROKS
Erasure Coded	27,775	23,049	+20.5%	ROKS
NFS Mid Tier	997.0	8,181	-87.8%	VCF
NFS Low Tier	496.0	4,086	-87.9%	VCF

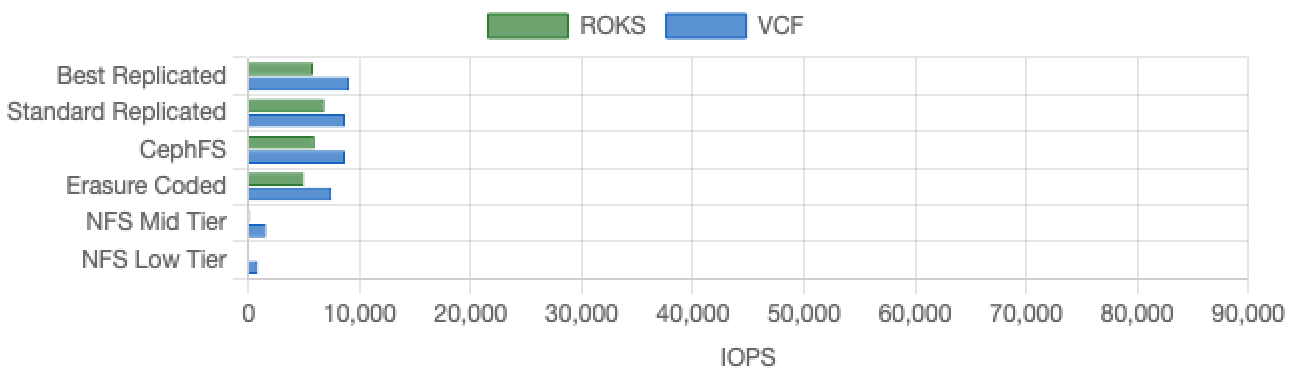
Avg p99 Latency (ms) — lower is better

Category	ROKS ROKS	VCF VCF	Delta	Winner
Best Replicated	5.6	10.3	-45.4%	ROKS
Standard Replicated	5.0	10.3	-51.7%	ROKS
CephFS	8.4	10.3	-18.4%	ROKS
Erasure Coded	49.0	13.3	+267.7%	VCF
NFS Mid Tier	150.0	18.4	+712.8%	VCF

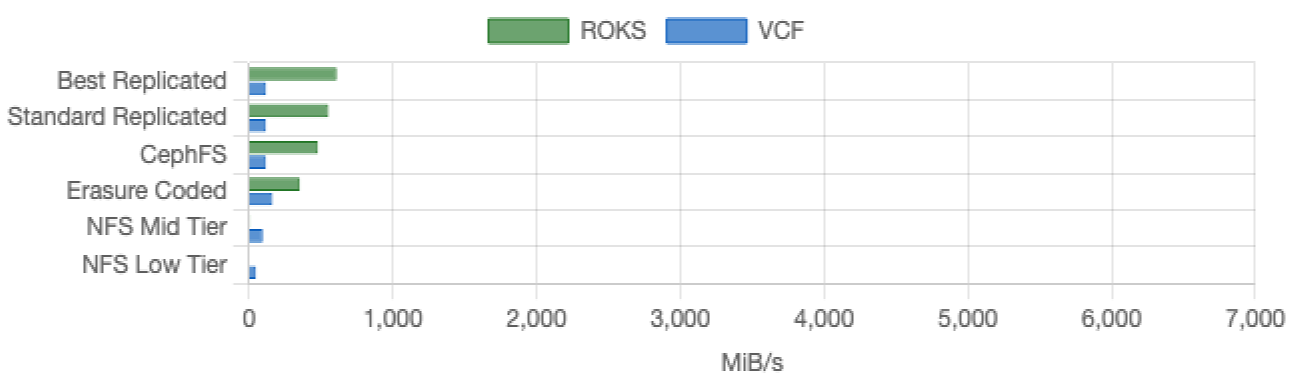
Category	ROKS ROKS	VCF VCF	Delta	Winner
NFS Low Tier	329.3	20.9	+1477.4%	VCF

Visual Comparison

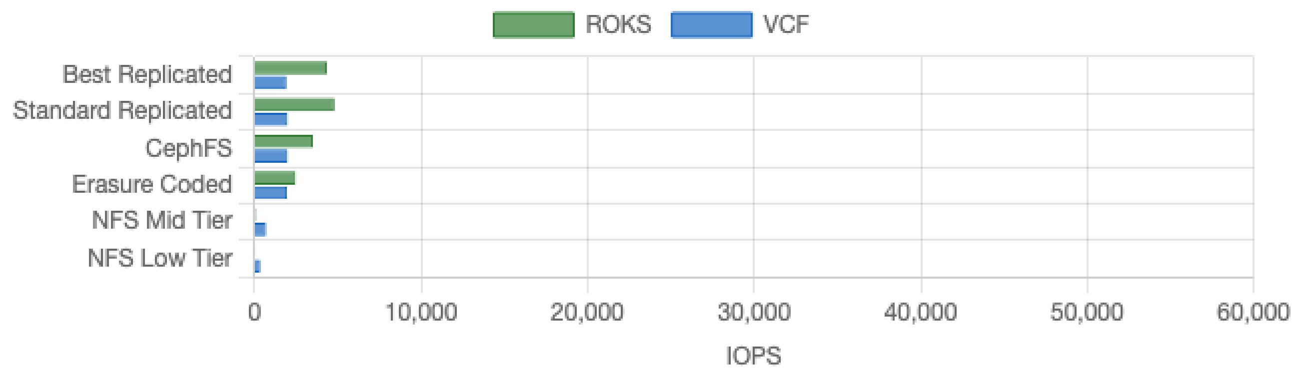
Random 4k IOPS (IOPS)



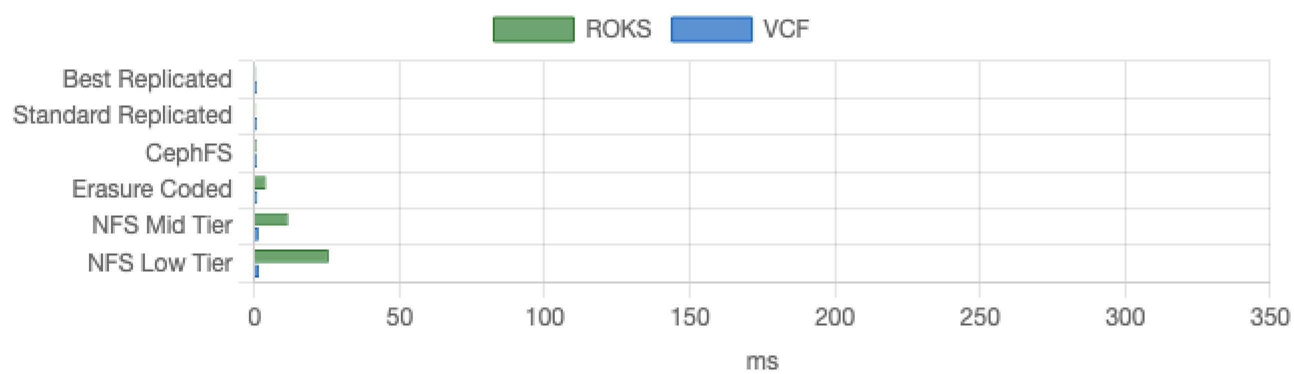
Sequential 1M BW (MiB/s)



Mixed 70/30 4k IOPS (IOPS)



Avg p99 Latency (ms) (lower is better)



Scorecard Summary

Category	IOPS Winner	Throughput Winner	Latency Winner
Best Replicated	VCF	ROKS	ROKS
Standard Replicated	VCF	ROKS	ROKS
CephFS	VCF	ROKS	ROKS
Erasure Coded	VCF	ROKS	VCF
NFS Mid Tier	VCF	VCF	VCF
NFS Low Tier	VCF	VCF	VCF

Key Takeaways

> **Methodology update (2026-05-26).** Earlier versions of this report cited ROKS write p99 latency in the 50–1,610 ms range. That was a measurement artifact — sparse-RBD image allocation cost in the first ~10 s of the 60 s measurement window plus reads from unallocated regions returning zeros. After the fix (prefill the test file, run fio against allocated data), real ROKS write p99 on replicated pools is 5–33 ms. VCF numbers are unchanged from 2026-02-25. The takeaways below reflect the corrected ROKS data.

> **ROKS still dominates sequential throughput.** ROKS delivered 3–5× the sequential read bandwidth of VCF on replicated block storage. Ceph's distributed architecture streams data across multiple OSDs in parallel, far exceeding what vSAN ESA delivers from local SSDs on a single host. This is ROKS's strongest advantage and survives the methodology fix.

> **ROKS wins mixed workloads convincingly.** With 70/30 read/write 4k I/O, ROKS delivered 1.2–2.3× the IOPS of VCF on block storage. Mixed write p99 on rep pools is now 5–6 ms — slightly *better* than VCF's reported 10 ms.

> **VCF still wins pure random IOPS on block storage.** With the corrected methodology, VCF leads ROKS by 24–36% on the read half of random-rw. The gap is real but smaller than the earlier (artifact-driven) numbers showed.

> **Tail latency on replicated pools is now competitive, not 4–5× worse.** The old "VCF wins on tail latency everywhere" claim was driven by the artifact. Real ROKS write p99 on rep2 for mixed workloads is 4.95 ms vs VCF's 10.25 ms — ROKS is actually slightly ahead for the most common VM workload pattern. VCF still leads on pure random writes by ~5–18 ms.

> **CephFS: strong throughput, modest random gap.** ROKS cephfs-rep3 (substituted for cephfs-rep2, which had a pre-existing auth-cap issue on the test cluster) delivered 4× the sequential throughput of VCF RAID-1 thin and 1.7× the mixed IOPS. Mixed write p99 is 8.36 ms — comparable to VCF's 10 ms, not 12× worse. The earlier 123 ms p99 number was the artifact.

> **Erasure coding: ROKS competitive on throughput, lags on random and latency.** ROKS ec-2-1 delivered +21% better mixed IOPS and +117% better sequential throughput than VCF RAID-5, while VCF led on random IOPS by 35% and on latency by ~7×. EC's write amplification (data + parity on every write) is real and harder to hide than the replication tax.

> **NFS high tier: not refreshed.** The earlier comparison's NFS high tier row used IBM Cloud Pool CSI (bench-pool). The Pool CSI driver isn't installed on the current test cluster, so this row is omitted from the updated charts. The old numbers are preserved in git history.

> **NFS mid/low tiers: VCF Endurance still wins decisively.** IBM Cloud File CSI's 500–1000 IOPS tiers are provisioned-IOPS-limited and don't scale with capacity. VCF's Endurance shares at 2–4 IOPS/GB scale linearly with disk size, delivering 7–8× better IOPS and throughput at these tiers. Architectural difference, not a methodology issue.